

Honors Precalculus Final Exam Formula Sheet

Cofunction Identities

$$\begin{aligned}\sin\left(\frac{\pi}{2} - u\right) &= \cos u & \cos\left(\frac{\pi}{2} - u\right) &= \sin u \\ \csc\left(\frac{\pi}{2} - u\right) &= \sec u & \sec\left(\frac{\pi}{2} - u\right) &= \csc u \\ \tan\left(\frac{\pi}{2} - u\right) &= \cot u & \cot\left(\frac{\pi}{2} - u\right) &= \tan u\end{aligned}$$

Pythagorean Identities

$$\begin{aligned}\sin^2 u + \cos^2 u &= 1 \\ 1 + \tan^2 u &= \sec^2 u \\ 1 + \cot^2 u &= \csc^2 u\end{aligned}$$

Double-Angle Formulas

$$\begin{aligned}\sin 2u &= 2 \sin u \cos u \\ \cos 2u &= \begin{cases} \cos^2 u - \sin^2 u \\ 2 \cos^2 u - 1 \\ 1 - 2 \sin^2 u \end{cases} \\ \tan 2u &= \frac{2 \tan u}{1 - \tan^2 u}\end{aligned}$$

Sum and Difference Identities

$$\begin{aligned}\sin(u \pm v) &= \sin u \cos v \pm \cos u \sin v \\ \cos(u \pm v) &= \cos u \cos v \mp \sin u \sin v \\ \tan(u \pm v) &= \frac{\tan u \pm \tan v}{1 \mp \tan u \tan v}\end{aligned}$$

Half-Angle Formulas

$$\begin{aligned}\sin \frac{u}{2} &= \pm \sqrt{\frac{1 - \cos u}{2}} \\ \cos \frac{u}{2} &= \pm \sqrt{\frac{1 + \cos u}{2}} \\ \tan \frac{u}{2} &= \frac{1 - \cos u}{\sin u} = \frac{\sin u}{1 + \cos u}\end{aligned}$$

Law of Cosines and Sines

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A \\ b^2 &= a^2 + c^2 - 2ac \cos B \\ c^2 &= a^2 + b^2 - 2ab \cos C\end{aligned}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Polar and Rectangular Coordinates; Trigonometric Form of a Complex Number

$$x = r \cos \theta \quad y = r \sin \theta \quad r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad x + yi = r(\cos \theta + i \sin \theta)$$

Sequences and Series

Arithmetic Sequence: $a_n = a_1 + d(n-1)$

Arithmetic Series: $S_n = \frac{n}{2}(a_1 + a_n)$

Geometric Sequence: $a_n = a_1 r^{n-1}$

Geometric Series (finite): $S_n = \frac{a_1(1 - r^n)}{1 - r}$

Infinite Geometric Series: $S_\infty = \frac{a_1}{1 - r}, \quad \text{if } |r| < 1$

Combination & Permutations

Combinations:

$${}_n C_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Permutations:

$${}_n P_r = \frac{n!}{(n-r)!}$$